

INTERFACING NOKIA COLOUR LCD WITH AVR MICROCONTROLLER

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Adding an elegant-looking colour LCD to a project is a dream for robotics enthusiasts. Usually, colour LCDs are costly and it's difficult to find the technical information for their interfacing.

Using the colour LCD of your old mobile phone could be a solution. Here we describe how to use

the colour LCD of a Nokia handset (model 6100, 7210, 6610, 7250 or 6220) with Philips PCF8833 chipset through ATmega2560 ATMEAL AVR microcontroller. These LCDs are readily available and inexpensive even if you buy a new one. The author's prototype is shown in Fig. 1. With this project, you can easily load a colour picture of 132×132 pixels, in 12-bit RGB (red-green-blue) format, to the colour LCD

through your PC's serial port.

ATmega2560 microcontroller

The ATmega2560 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. The AVR core combines a rich instruction set with 32 general-purpose working registers. All the 32 registers are directly connected to the arithmetic logic unit, allowing two independent registers to be accessed in one single instruction executed in one clock cycle. The resulting architecture is more code-efficient while achieving throughputs up to ten times faster than conventional CISC microcontrollers.

The microcontroller has 256 kB of in-system programmable Flash with read-while-write capabilities, 4kB EEPROM, 8kB SRAM, 86 general-purpose input/output (I/O) lines, real-time counter, six flexible timers/counters with compare modes and pulse-width modulation (PWM), four USARTs,

a byte-oriented two-wire serial interface, a 16-channel, 10-bit analogue-to-digital converter (ADC) with optional differential input stage with programmable gain, programmable watchdog timer with internal oscillator, a serial peripheral interface (SPI) port, IEEE standard 1149.1 compli-



Fig. 1: Author's prototype

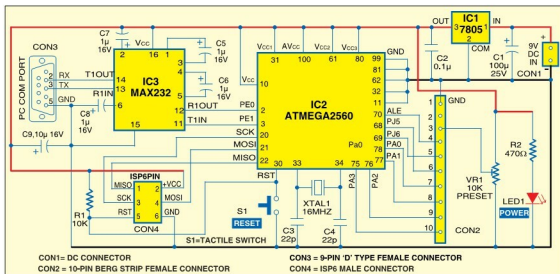


Fig. 2: Circuit for interfacing the Nokia colour LCD with AVR microcontroller

